
ME 638: Solar Thermal Power Plant

Course Code	ME 638
Course Name	Solar Thermal Power Plant
L-T-P-C	3-0-0-3
Prerequisites	Instructor Consent
Intended for	UG/M. Tech. /MS/PhD
Distribution	Specialized stream elective course for M. Tech. in Mechanical Engineering with specialization in Energy Systems, and elective course for other students
Approval	10th Senate (Jun 2016)

Course Contents

- **Module I**

Physical processes that determine the output of a solar thermal collector, and mathematical models that can be used to calculate this output, relation between factors and models, materials and surface treatments for use in a solar collector, optical parameters of absorbers, reflectors and transparent materials and comparison of their characteristics. [6 Lectures]

- **Module II**

Evaluate of different thermal loads and its mathematical modeling, design of different solar thermal systems and comparing their suitability for different boundary conditions [8 Lectures]

- **Module III**

Calculation of storage capacity for different heat storage techniques and determine their suitability for given boundary conditions. [7 Lectures]

- **Module IV**

Function of the important components necessary in a solar thermal system and the testing standards used for their evaluation. [7 Lectures]

- **Module V**

Need and challenges involved in using solar energy for cooking, cooling, distillation and drying applications as well as identify niche areas for these applications, functioning of different designs and system dimensions of solar cookers, stoves, cooling systems and dryers [14 Lectures]

Text Books

1. Peter Gevorkian, **Large-Scale Solar Power System Design: An Engineering Guide for Grid Connected Solar Power Generation**, McGraw-Hill, 2011.